Chained Operations: (a + b) × (c - d) on the Stack

Numerical & Algebraic Expressions • Standard: CCSS.MATH.6.EE.A.2

1. MATHEMATICAL CONCEPT & STACK APPROACH: Map an algebraic expression to a postfix sequence and execute on the 4-register stack without parentheses.

Written algebra uses parentheses to show which inner operations must happen first. Postfix (stack) notation evaluates expressions in true mathematical order from the inside out: compute the first branch (6 + 1 = 7), let it wait on Y while computing the second branch (9 - 5 = 4), and multiply.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28.

Student Prediction (Before Calculator): Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [6] [ENTER] [1] [+] [9] [ENTER] [5] [-] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[6]	6	Value a (6) placed into Register X
[ENTER]	6	ENTER lifts 6 into Register Y
[1]	1	Value b (1) placed into Register X
[+]	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
[9]	9	Value c (9) entered into X; 7 lifted to Y
[ENTER]	9	ENTER lifts 9 into Z and 7 into T
[5]	5	Value d (5) placed into Register X
[-]	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
[×]	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

4. STORY CONTEXT (Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1961 Railway Siding)

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the Prime Factor Studio (where 24 = 2 × 2 × 2 × 3 decomposes into prime factor towers), students evaluate nested expressions from the inside out.

Core Investigation (Student Warm-Up A):

Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Deep Stack Evaluation & Register Limits

Order of Operations & Stack Evaluation • Standard: CCSS.MATH.6.EE.A.2

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Deeply nested fractions such as (12 - 4) ÷ (2 × (1 + 1)) require holding completed sub-expressions aside while inner terms are resolved. The 4-register stack (T, Z, Y, X) holds up to four nested levels visibly without memory variables or parentheses.

Blocks & Prediction: Track the 4 stack registers (T, Z, Y, X) alongside Prime Factor towers for (12 - 4) / (2 × (1 + 1)). Prediction: Numerator: 12 - 4 = 8 (three [2] blocks). Denominator: 2 × 2 = 4 (two [2] blocks). Result: 8 / 4 = 2.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	First digit of 12
[2]	12	12 in register X
[ENTER]	12	12 lifted to Y
[4]	4	4 in register X
[-]	8	Numerator 12 - 4 = 8 held in X
[2]	2	Factor 2 enters X; 8 lifts to Y
[ENTER]	2	2 lifted to Y; 8 lifts to Z
[1]	1	Inner 1 enters X
[ENTER]	1	1 lifted to Y; 2 in Z; 8 in T
[1]	1	Second 1 enters X
[+]	2	Innermost sum: 1 + 1 = 2 in X
[×]	4	Denominator combined: 2 × 2 = 4 in X; 8 in Y
[÷]	2	8 / 4 = 2!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.

Hands-On Block Remediation: Model the product branch 5 × 2 with Prime Factor [5] and [2] blocks first: push 10 to Register Y ([10] [ENTER]), evaluate 5 × 2 in X ([5] [ENTER] [2] [×] → 10), then press [+] → 20.

Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).

Hands-On Block Remediation: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

4. EXTENSION MATHEMATICAL INSIGHT

Deeply nested fractions such as (12 - 4) ÷ (2 × (1 + 1)) require holding completed sub-expressions aside while inner terms are resolved. The 4-register stack (T, Z, Y, X) holds up to four nested levels visibly without memory variables or parentheses.

Postfix Evaluation, Factor Trees & Chained Operations

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: POSTFIX EVALUATION & PRIME FACTOR TREE RULES

- In postfix evaluation, operands are entered first, followed immediately by operators.
- Evaluation order matches mathematical precedence: evaluate inner sub-expressions and factor branches first.
- RPN keystroke sequence evaluates complex expressions naturally without parenthesis keys.

Key Sequence: `6` `ENTER` `1` `+` `9` `ENTER` `5` `-` `×`

Investigation 1: Evaluate (4 + 3) × (10 - 6) and check on the stack.

1. Block Model / Fraction Sketch:

2. Paper Work & Stack Steps:

Math Equation / Setup:

Two Shelves (Y, X):

Stack Steps:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Solve (20 - 4) / (2 × 4) using inner-first evaluation.

1. Block Model / Fraction Sketch:

2. Paper Work & Stack Steps:

Math Equation / Setup:

Two Shelves (Y, X):

Stack Steps:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack.

1. Block Model / Fraction Sketch:

2. Paper Work & Stack Steps:

Math Equation / Setup:

Two Shelves (Y, X):

Stack Steps:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product first.

1. Block Model / Fraction Sketch:

2. Paper Work & Stack Steps:

Math Equation / Setup:

Two Shelves (Y, X):

Stack Steps:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 07: Postfix Evaluation, Factor Trees & Chained Operations — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 5–6

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Core Solution: 50. Keystrokes: [8] [ENTER] [2] [+] [1][4] [ENTER] [9] [-] [×]

Core Check Keys: `8` `ENTER` `2` `+` `1` `4` `ENTER` `9` `-` `×`

Extension Challenge (Warm-Up B): Evaluate (15 - 5) / (3 + 2) by computing the numerator and denominator, then verifying on the stack.

Extension Solution: 2. Keystrokes: [1][5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Ext. Check Keys: `1` `5` `ENTER` `5` `-` `3` `ENTER` `2` `+` `÷`

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Evaluate (4 + 3) × (10 - 6) and check on the stack.

Expected Answer: 28

SC-6 Verification Keys: `4` `ENTER` `3` `+` `1` `0` `ENTER` `6` `-` `×`

Investigation 2: Solve (20 - 4) / (2 × 4) using inner-first evaluation.

Expected Answer: 2

SC-6 Verification Keys: `2` `0` `ENTER` `4` `-` `2` `ENTER` `4` `×` `÷`

Investigation 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack.

Expected Answer: 4

SC-6 Verification Keys: `1` `8` `ENTER` `6` `-` `1` `ENTER` `2` `+` `÷`

Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product first.

Expected Answer: 17

SC-6 Verification Keys: `3` `ENTER` `4` `×` `5` `+`

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.